

Attribution Platform: Product Scope

Working name: Wink Attribution (rename later). Version 1.0 of the scope, October 2026.

1. What we are building

A multi-tenant platform that tells a company which marketing source brought each paying customer or booking, and shows the results as accurate revenue, customer and booking reports.

A company connects its website and its payment, booking or e-commerce system. The platform records every visit with its source, carries a small token through checkout, receives the payment or booking afterward, and credits the right channels.

It must work for any business, anywhere: a Zenoti salon, a Shopify store, or a custom-built system. The first target niche is service businesses with bookings, where upcoming bookings and booked value matter.

Goals

1. Show the true source of every sale and booking, including journeys with several sources (Meta, then Google, then purchase).
2. Make numbers accurate and explainable: every figure can be traced to the orders behind it.
3. Let a new company go live in under 30 minutes without our help.
4. Run globally: any timezone, any currency, per-country privacy rules.
5. Keep each company's data fully isolated, with a platform admin who can support all of them.

Non-goals for version 1

- Sending conversions back to Google or Meta for ad optimization (kept as an opt-in later phase).
- Ad creation, bidding or budget management.
- Predictive models, probabilistic or cross-device fingerprinting.
- Native mobile app tracking.
- A white-label reseller portal.

2. People and roles

Role	Belongs to	Can do
Platform admin	No company (our team)	See and manage all companies, approve onboarding, suspend, support access (read-only with reason), view platform health
Company admin	One company	Manage users, sites, locations, API keys, connectors, attribution settings, billing; see all company data
Company viewer	One company	Read-only dashboard for the company, optionally limited to certain locations
Machine client	One company	Calls the public API using a key (publishable or secret)

Rule: every query is scoped by company, taken from the login or API key and never from the request

body.

3. How it works end to end

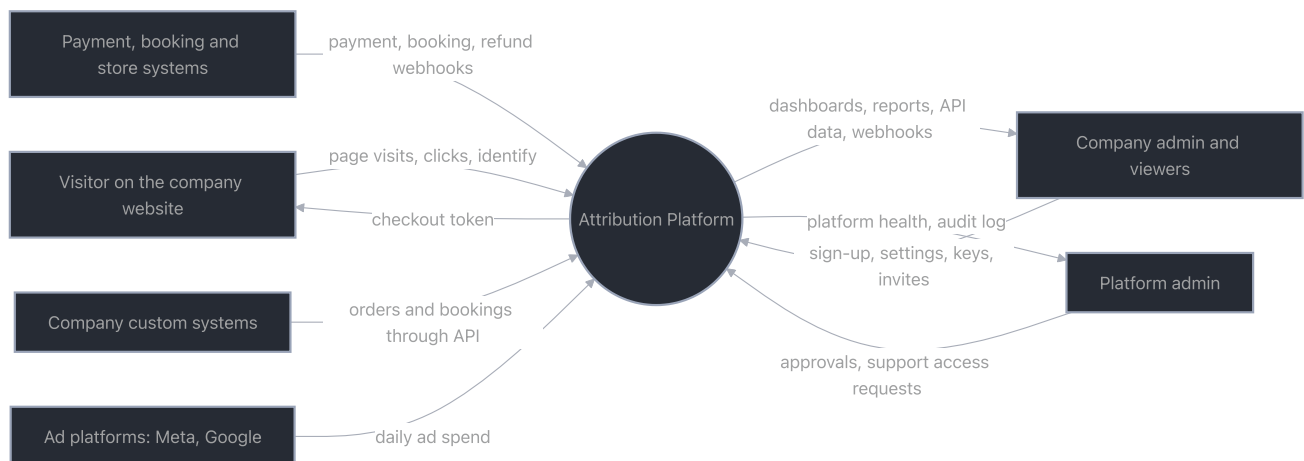
1. The Site SDK records each visit as a touch: source, medium, campaign, click IDs, referrer, landing page, time.
2. When the visitor starts checkout or booking, the site requests a checkout token that freezes that visitor's touches.
3. The site puts the token into the payment or booking system's extra field (Stripe metadata, Razorpay notes, PayPal custom ID, Shopify cart attributes, WooCommerce order meta, Zenoti notes or UTM fields).
4. When payment or booking completes, the system's webhook or the company's server sends it to us with the token.
5. We match the token to the visitor's touches, resolve the customer, and compute credit under each attribution model.
6. Dashboards and the API show the results.
7. If no token exists, we fall back to matching by hashed email or phone, then by click ID, and label the match confidence.

3.1 Data flow diagrams (DFD)

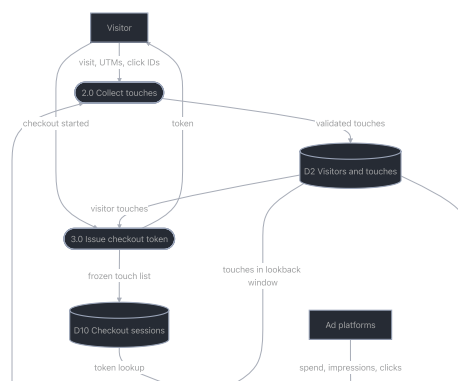
Notation: rectangles are external parties, rounded shapes are processes, cylinders are data stores, and arrows are labelled with the data that moves.

Level 0: context diagram

The whole platform as one process, and everything that talks to it.



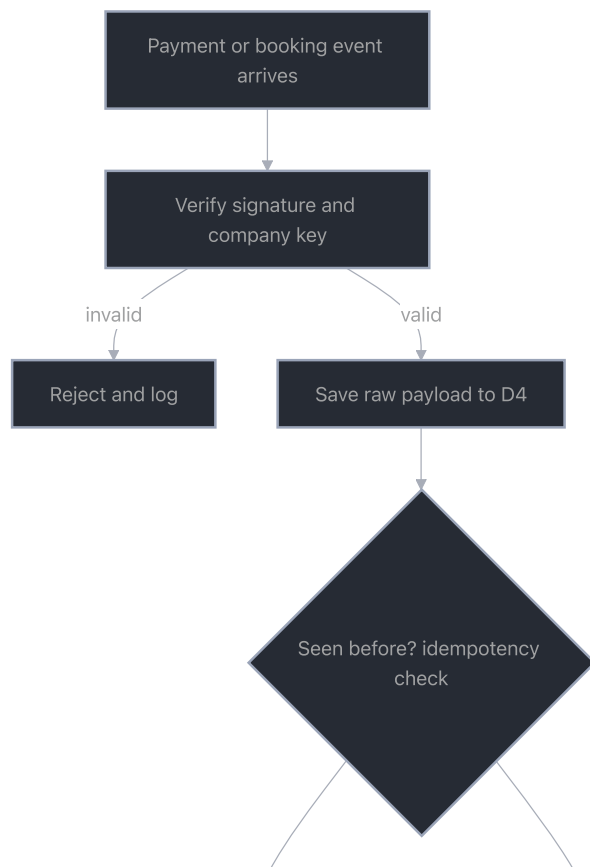
Level 1: main processes and data stores

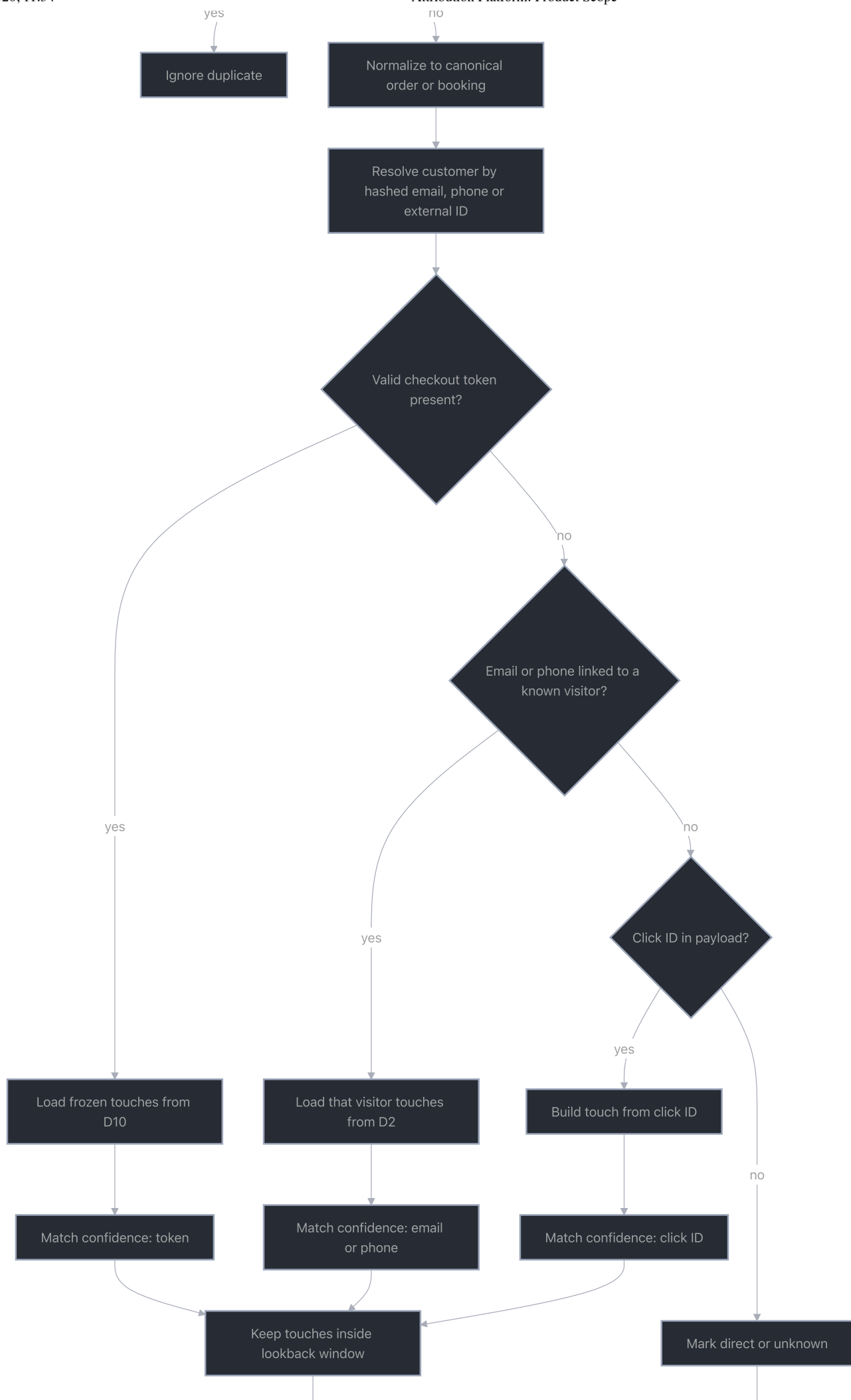


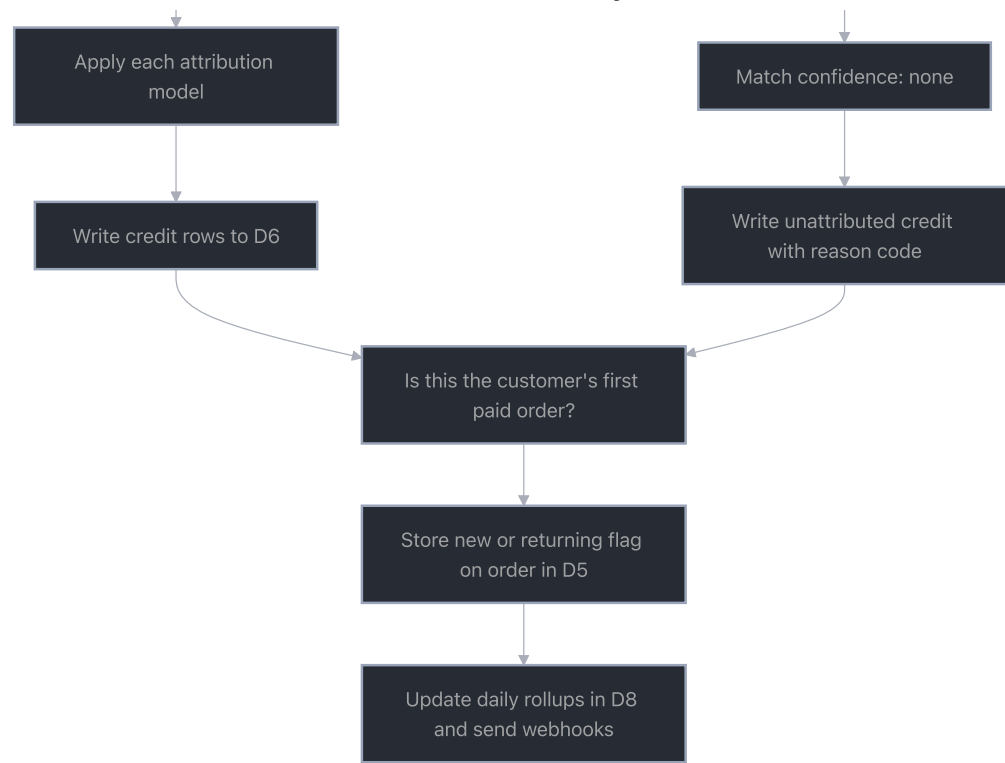


Level 2: how a payment is matched to its source (processes 4.0, 5.0, 6.0)

This is the most important flow in the product, so it is shown step by step, including the fallbacks and the confidence label.







Data store dictionary

Store	Holds	Written by	Read by
D1 Workspaces, users, keys	Companies, settings, users, hashed API keys, connector config	1.0, 10.0	2.0, 4.0, 6.0, 7.0, 9.0
D2 Visitors and touches	Visitor IDs, every touch with source and click IDs	2.0	3.0, 5.0, 6.0
D3 Customers	Customer records, hashed email and phone, external IDs, merges	5.0	5.0, 9.0
D4 Raw events	Original payloads for replay	4.0	Replay tool, 10.0
D5 Orders and bookings	Canonical orders, refunds, bookings, new or returning flag	4.0, 5.0	6.0, 7.0, 9.0
D6 Attribution credits	Credit share per order, touch and model	6.0	7.0, 9.0
D7 Ad spend	Daily spend by campaign	8.0	7.0
D8 Metric rollups	Daily metrics by location, channel, campaign, model	7.0	9.0
D9 Audit log	Logins, changes, platform admin access	1.0, 10.0	10.0, 9.0
D10 Checkout sessions	Token and frozen touch list	3.0	6.0

Trust boundaries

1. Browser to Edge API: untrusted. Only the publishable key (write-only, origin restricted) is accepted, and all input is validated.
2. Payment and booking systems to Edge API: signature or secret verified before anything is trusted.
3. Company to company: never crosses. Every read and write on D1 to D10 carries the company ID taken from the key or login.
4. Platform admin to company data: read-only, reason required, and always written to D9.

4. Scope by module

4.1 Onboarding and company management

Must have (MVP):

- Sign-up form: company name, admin name and email, country, timezone, reporting currency, website URL, industry.
- Email verification.
- Workspace status: pending, active, suspended, closed.
- Domain verification by DNS TXT record or meta tag. API keys and SDK stay inactive until verified.
- Optional manual approval by the platform admin before going live (configurable).
- Setup checklist inside the portal: verify domain, create keys, install SDK, connect system, send test event, go live.
- Test event tool: sends a sample event and shows it arriving in a live feed, with errors explained in plain language.
- Invite users by email with role selection.
- Company settings: name, timezone, week start day, currency, attribution model default, lookback window, channel mapping, consent mode.
- Locations: name, timezone, currency, external IDs (for example Zenoti center ID).

Later:

- Self-serve billing and plans, usage limits, invoices.
- Single sign-on for larger companies.
- Agency level above companies (one login managing many clients).

4.2 Site SDK (JavaScript tracker)

Must have:

- One script tag, under 10 KB, loaded asynchronously.
- Creates a first-party visitor ID.
- Captures UTM parameters, fbclid, gclid, ttclid, msclkid, affiliate click IDs, referrer, landing URL, page title and timestamp.
- Captures the _fbclid, _fbp and _ga cookies when present.
- Sends touches to the API with retry and queueing.
- Methods: identify(email, phone), getCheckoutToken(), decorateLinks() (adds the token to outbound booking-widget links), consent(granted).
- Writes the token into forms and carts automatically for known platforms (Shopify, WooCommerce) and through a documented call for custom sites.

- Respects consent: collects nothing before consent where required.
- Works inside Facebook and Instagram in-app browsers and handles Safari's cookie limits as well as possible.

Later:

- First-party subdomain mode (CNAME) for longer cookie life.
- Single-page-app helpers for React, Vue and Next.js.

4.3 Public API

Authentication: publishable key (browser, write-only, restricted to allowed origins) and secret key (server). Keys are stored hashed, can be rotated, and have per-key rate limits.

Write endpoints:

- `POST /v1/touches`
- `POST /v1/identify`
- `POST /v1/checkout-sessions` returns the token
- `POST /v1/orders` (or `/v1/conversions`)
- `POST /v1/bookings` and `PATCH /v1/bookings/:id`
- `POST /v1/refunds`
- Every write accepts an idempotency key.

Read endpoints:

- `GET /v1/orders`, `GET /v1/orders/:id/attribution` (journey plus credit under every model)
- `GET /v1/customers`, `GET /v1/customers/:id/journey`
- `GET /v1/bookings`, `GET /v1/bookings/upcoming`
- `GET /v1/reports/summary`, `/channels`, `/campaigns`, `/locations`, `/trend`, `/customers`, `/cac`
- All reports accept date range, tz, model, location, channel and currency.

Other:

- Outbound webhooks (order attributed, booking created, integration failing), signed with HMAC and retried.
- Versioned API, cursor pagination, consistent error format, public documentation with copy-paste examples.
- A sandbox mode with test keys.

4.4 Connectors (inbound integrations)

Every connector converts the source's payload into one canonical event: `touch`, `identify`, `booking.created`, `booking.updated`, `booking.cancelled`, `order.paid`, `order.refunded`.

Connector	Phase	Notes
Zenoti	MVP	Repackage the existing handlers: guest, appointment, invoice events

Stripe	MVP	Webhook on payment success and refund; token from metadata
Shopify	MVP	App or webhook; token from cart attributes
Razorpay	MVP	Webhook; token from notes
Generic REST and webhook with field mapping	MVP	Lets a custom system integrate with no code from us
WooCommerce, PayPal, Square	Phase 2	
Other booking systems (Fresha, Mindbody, Vagaro)	Phase 3	

Connector requirements: raw payload stored before processing, replay on failure, signature verification, handling of duplicates and out-of-order events, and a health status per connector (last event received, error count).

4.5 Identity resolution

- Anonymous visitor to known customer using email or phone (stored hashed for matching, with the original kept only where needed).
- Customer merge handling: when two customers are merged in the source system, their touches and orders combine and attribution is recomputed.
- Multiple visitor IDs per customer (phone and laptop) once the person identifies themselves.
- Match confidence recorded on every conversion: token, click ID, email or phone, or none.
- Deterministic rules only in version 1.

4.6 Attribution engine

- Stores every touch; never overwrites.
- Models: first touch, last touch, last non-direct touch (default), linear, time decay, position based (40/20/40).
- Per-company default model, lookback window (default 30 days, configurable per channel type) and channel mapping rules.
- Touch priority: click-ID touch, then UTM touch, then referrer-only, then direct. A later direct visit never replaces a paid click.
- Writes `AttributionCredit` rows (order, touch, model, share, amount) so any model can be shown instantly.
- Re-run tool: recompute history when settings or logic change.
- Bookings are attributed when created, not when paid; the credit carries forward when the invoice is paid.
- Unattributed orders are shown as "Direct / unknown" with a reason code.

4.7 Metrics engine

Precomputed rollups by company, day (in company timezone), location, channel, campaign, model and new-versus-returning. Rebuilt automatically when timezone, currency or model settings change.

Exact definitions are in section 6.

4.8 Ad spend and CAC

- Connect Meta and Google Ads accounts by OAuth to pull daily spend, impressions and clicks by campaign.
- Store the ad account's timezone and currency and align them to the company's before computing CAC.
- Manual spend upload (CSV) for other channels.
- Reports: blended CAC, CAC per channel, ROAS, cost per booking, payback view.

4.9 Platform admin console

- Company list with status, plan, country, events per day, last event time, integration health.
- Approve, suspend or reactivate a company.
- Open any company's dashboard in read-only mode, with a required reason, masked personal data by default, and every access written to an audit log that the company can also see.
- Platform health: events per minute, queue depth, failed events, slowest connectors, errors by company.
- Tools: replay events, re-run attribution, rotate a key, delete a company's data on request.
- Two-factor login required.

4.10 Security and privacy

- Strict per-company isolation enforced in one data-access layer, with automated tests that try to read across companies.
- Hashed API keys, origin allow-lists, rate limits, signed webhooks, secrets encrypted at rest.
- Personal data: email and phone hashed for matching; masking in UI; data export and deletion by customer; retention settings (for example raw payloads 13 months).
- Consent mode per company; support for GDPR, UK GDPR, DPDP (India), CCPA.
- Audit log for logins, key creation, settings changes, admin access.
- No logging of full payloads or personal data in application logs (the current system logs raw bodies; this must stop).
- Password policy, two-factor option, session expiry, login throttling.

4.11 Time, currency and language

- All timestamps stored in UTC. Company timezone and optional per-location timezone use IANA names.
- Daily, weekly and monthly groupings use the company's local day. Week start is configurable.
- Money stored as integer minor units plus currency code. A daily FX rate snapshot is kept so reports in the reporting currency are reproducible.
- Interface language: English at launch, with text structured for translation.

4.12 Reliability and operations

- Event acknowledgement within 200 ms at the edge; processing asynchronous.
- At-least-once processing with idempotency so duplicates never double-count.

• Send letterpress and a dashboard for failed events

- Dead-letter queue and a dashboard for failed events.
- Targets: 99.9 percent API availability, 95 percent of events processed within 60 seconds, zero lost accepted events.
- Daily backups, tested restore, environment separation (dev, staging, production), monitoring and alerting.

5. What the dashboard must show

All screens respect the company timezone, currency, location filter, date range, attribution model and channel filter. Every number can be clicked to open the list of orders behind it. Every screen has an Export (CSV) action and a "last updated" time.

5.1 Overview

- Net revenue, orders, average order value, new customers, returning customers, repeat rate.
- Each with change versus the previous period.
- Revenue by channel (ledger bar), trend chart by day with the previous period overlaid.
- Top campaigns, top locations.
- Attribution coverage: percent of revenue with a known source, and how it was matched (token, click ID, email or phone, none).
- Alerts: connector failing, no events in 24 hours, coverage dropped.

5.2 Channels and campaigns

- Table by channel: revenue, orders, new customers, spend, CAC, ROAS, average order value, average days to convert.
- Drill into a channel to see campaigns, ad sets and ads where available.
- Model switcher (first, last, last non-direct, linear, time decay, position based), with a comparison view showing the same channel under each model side by side.
- Drill into a campaign to list its customers and orders.

5.3 Customer journeys

- A path report: the most common sequences ("Meta then Google then purchase"), with count and revenue.
- Average number of touches and days from first touch to purchase.
- Assisted conversions: channels that appear in a journey but are not the last touch.
- A customer page with the full timeline: touches, bookings, orders, refunds, and the credit given to each source.

5.4 Orders and transactions

- Every order with date, customer, location, services or items, gross, discount, tax, net, refund, source under the selected model, match confidence, new or returning.
- Filters: location, channel, status, new vs returning, refunded, unattributed.
- Order detail with the journey and each model's credit.
- Refunds listed separately and linked to the original order.

5.5 Customers

- Searchable, sortable list: first seen, first purchase, last purchase, orders, lifetime revenue, first source, last source.
- New versus returning split with trend.
- Cohort view: customers by first-purchase month, with repeat purchase rate over time.
- Lifetime value by acquisition channel.

5.6 Bookings and upcoming

- Upcoming bookings: count, booked value, by day and week in local time, by location, by service, by source.
- New guest versus returning guest on every booking.
- Status: scheduled, confirmed, completed, cancelled, rescheduled, no-show.
- Booked value (service price at booking) shown separately from realized revenue (paid).
- Cancellation rate, no-show rate and average lead time (booking date to appointment date), by channel.
- Pipeline view: bookings created in the period and how many converted to completed and paid.

5.7 Locations

- Revenue, orders, new and returning customers, bookings, top sources and CAC per location.
- Location comparison table and ranking.
- Each location uses its own timezone for day boundaries.

5.8 Ad spend and CAC

- Spend by platform and campaign, with impressions and clicks.
- Blended CAC and CAC by channel, ROAS, cost per booking.
- Spend versus revenue trend.
- Warnings when ad-account timezone or currency differs from the company setting.

5.9 Integrations and data quality

- Connector status: last event, events today, error count, retry button.
- Event feed (live) with the raw payload visible to admins only.
- Data quality checks: orders with no token, duplicate customers, missing currency, amounts that look wrong, events out of order.
- Replay failed events.

5.10 Settings

- Company profile, timezone, currency, week start, default model, lookback window, channel mapping rules, consent mode.
- Team: invite, change role, limit by location, deactivate.
- API keys: create, rotate, revoke, view last use, allowed origins.
- Webhooks: endpoints, event types, delivery log.
- Audit log: logins, changes and platform admin access.
- Data: export, delete a customer, retention settings.

6. Metric definitions

Every metric needs one written definition, used identically in the dashboard, API and exports.

Metric	Definition
Gross revenue	Sum of paid order totals before discounts and refunds, excluding tax
Discounts	Total discounts applied to paid orders
Refunds	Sum of refund amounts, counted on the refund date, linked to the original order
Net revenue	Gross revenue minus discounts minus refunds
Orders	Count of paid orders (fully paid invoices; unpaid or voided excluded)
Average order value	Net revenue divided by orders
New customer	Customer whose paid order is their first paid order in the company, judged by the order timestamp after identity resolution
Returning customer	Customer with at least one earlier paid order
New customer revenue	Net revenue from orders by new customers
Repeat rate	Customers with 2 or more paid orders divided by customers with 1 or more, in the period
Attributed revenue	Net revenue multiplied by the credit share for the selected model
Attribution coverage	Revenue with a matched source divided by total revenue
Booked value	Sum of service prices on bookings with status scheduled or confirmed
Realized revenue	Net revenue from paid orders
Cancellation rate	Cancelled bookings divided by created bookings in the period
No-show rate	No-show bookings divided by past bookings that were scheduled
Lead time	Appointment start minus booking created time
Days to convert	First touch time to order time
CAC	Ad spend divided by new customers (blended and per channel)
ROAS	Attributed revenue divided by ad spend
Cost per booking	Ad spend divided by bookings created

Rules that apply everywhere:

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- Attribution Platform: Product Scope
- New versus returning is decided when the order is ingested and stored on the order. It is recomputed only on a customer merge or a backfill.
 - A "day" is the company's local day. Last 7 days means the 7 local days ending today.
 - Refunds reduce revenue on the date of the refund, with a toggle to show them against the original sale date.
 - Gift cards, packages and memberships need a defined treatment: revenue is counted when the service or product is paid for, and package redemptions are not counted as new revenue.
 - Tax is excluded from revenue unless the company chooses otherwise.
 - Amounts are converted to the reporting currency at the rate stored for the order date.

7. Main data entities

- Workspace: name, status, region, timezone, currency, week start, settings, verified domains.
- Site: workspace, domain, allowed origins, SDK settings.
- Location: workspace, name, timezone, currency, external IDs.
- User: workspace (empty for platform admins), role, location access, two-factor.
- Invite, ApiKey, WebhookEndpoint, AuditLog.
- Visitor: workspace, visitor ID, first seen, linked customer.
- Touch: workspace, visitor, time, channel, source, medium, campaign, click IDs, referrer, landing URL.
- Customer: workspace, hashed email and phone, external IDs map, first seen, first purchase, merged-into.
- CheckoutSession: token, visitor, frozen touch list, expiry, consumed order.
- Order: workspace, location, customer, external ID, amounts in minor units, currency, FX snapshot, status, is-new-customer, token, match method.
- OrderItem, Refund.
- Booking: workspace, location, customer, service, appointment time (UTC), status history, booked value, token, source.
- AttributionCredit: order or booking, touch, model, share, amount.
- AdAccount, AdSpend (daily, by campaign), FxRate.
- RawEvent: source, payload, received time, status, error, replay count.
- Rollup tables for daily metrics.

8. Build phases

Phase	Contents	Outcome
0. Foundation	Security cleanup of the current system; rename Brand to Workspace; hashed API keys; per-company scoping; UTC and timezone utilities; canonical event format	Safe base to build on
1. Core	SDK, touches, checkout tokens, orders and bookings API, Zenoti, Stripe, Shopify, Razorpay and generic connectors, identity, first and last touch, event	A company can connect and see correct attributed

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	queue, replay	orders
2. Reports	Metrics engine with rollups, the dashboard screens in section 5, new vs returning, upcoming bookings, refunds, exports	Numbers a business can trust
3. Onboarding and admin	Self-serve onboarding, domain verification, setup checklist, test events, platform admin console, audit log	Companies join without us
4. Depth	All attribution models, journeys and cohorts, ad spend and CAC, outbound webhooks, data quality screen	Full analytics
5. Scale	More connectors, outbound conversions to ad platforms (opt-in), regional deployments, billing, single sign-on, agency level	Growth

9. Acceptance criteria (how we know it is accurate)

- 1. A test journey of Meta click, Google click, then purchase appears with first touch Meta, last touch Google, and linear 50/50.
- 2. The same payment sent twice creates one order.
- 3. A refund reduces net revenue on its date and is linked to the original order.
- 4. A customer merge combines orders and recalculates new versus returning.
- 5. Revenue for "today" matches the company's local midnight to midnight, including across a daylight saving change.
- 6. A user from company A can never read company B's data (automated test).
- 7. Dashboard totals equal the sum of the orders shown when drilled down.
- 8. A platform admin's data access appears in the company's audit log.
- 9. Replaying a day of raw events produces identical results.
- 10. Attribution coverage is displayed and matches a manual sample check to within an agreed tolerance.

10. Risks and open decisions

Risks:

- Ad blockers, Safari cookie limits, in-app browsers and cross-device journeys always lose some touches. We state coverage openly.
- Gateways differ in how much extra data they pass back, so some connectors will have lower match rates.
- Privacy laws require consent handling and deletion tools from day one.